Week #7: [Non-Mechanical Back & Neck Pain]

(Non-hyperlinked = Dr. Caitlin Cassidy Module)

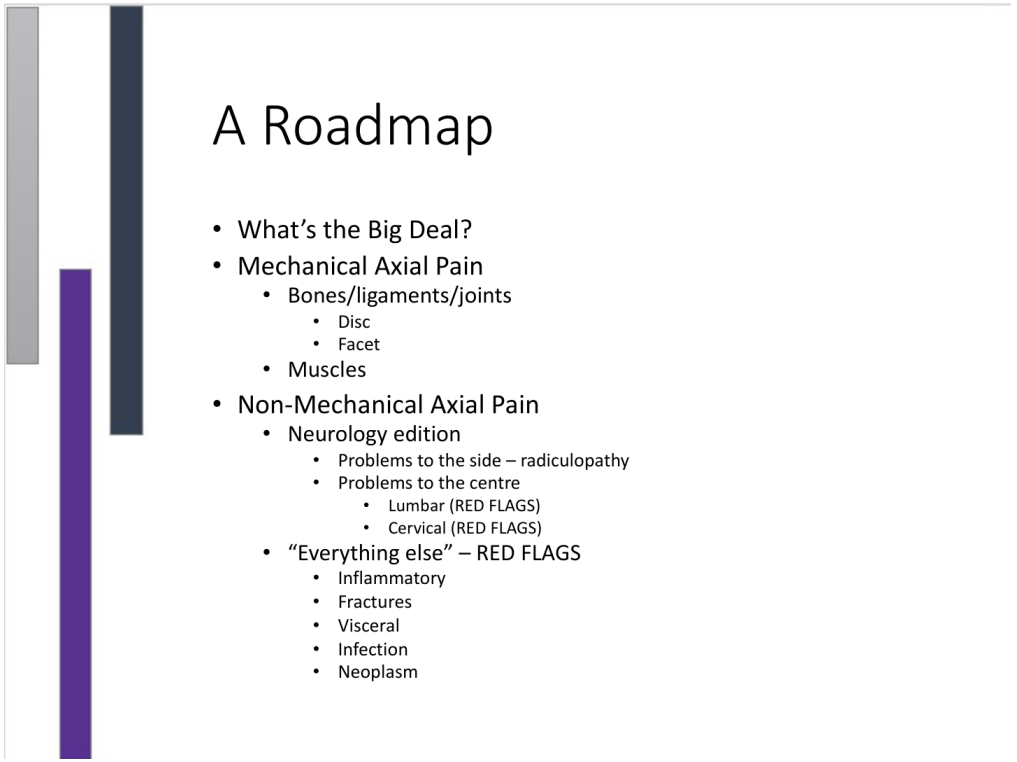
- Introduction to Module

Non-Mechanical Spine Pain: Neurology Edition

- **Anatomy of the Spinal Cord (32 min)**
- Non-Mechanical Spine Pain: Neurology Edition (20 min)

Non-Mechanical Spine Pain: 'Everything Else'

- Non- Mechanical Spine Pain: RED FLAGS (9 min)



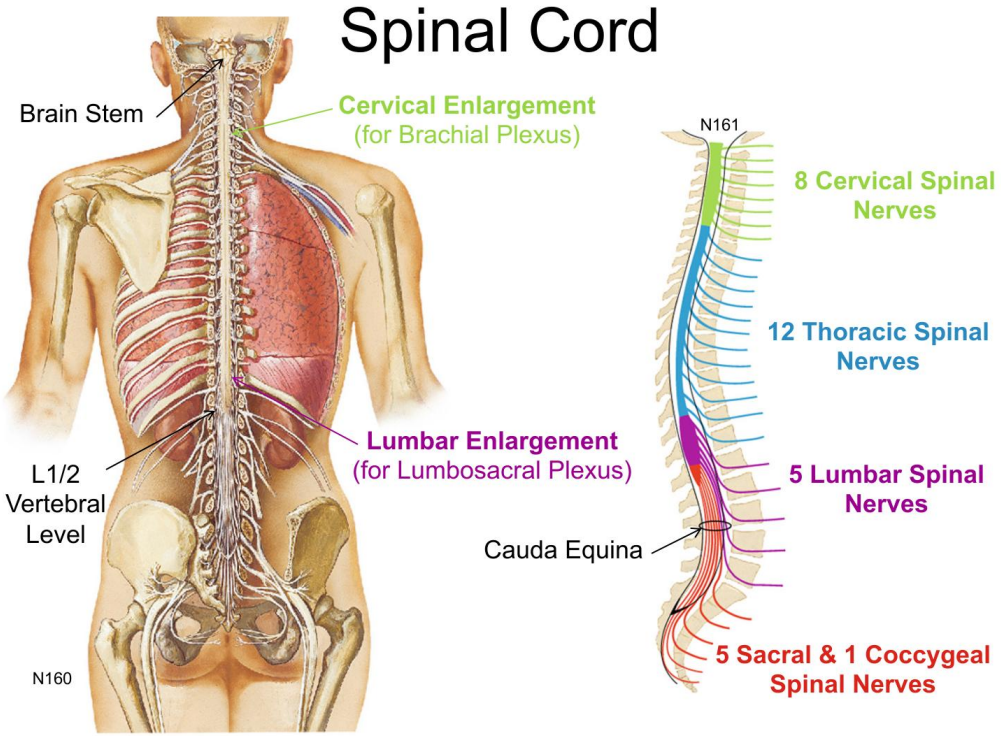
Non-Mechanical Spine Pain (Neurology Edition)

Anatomy of the Spinal Cord

Non-Mechanical Spine Pain (Neurology Edition)

- We will be using disc herniation to illustrate neurologic back pain because it can occur as a natural extension of degenerative disc disease (discussed in the previous videos) AND because herniation/spondylosis are the most common causes of neurologic back pain BUT
  - Neurologic compromise comes from ANYTHING squishing the nerves, so have to have a differential for this (abscess, tumour, bleed, subluxation/dislocation (ie. spondylolisthesis), etc).
- Let's start with Anatomy!!

Anatomy of the Spinal Cord



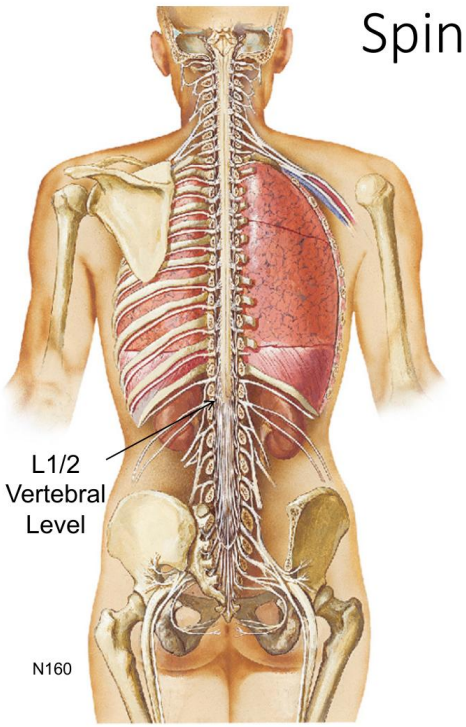
Anatomy of the Spinal Cord:

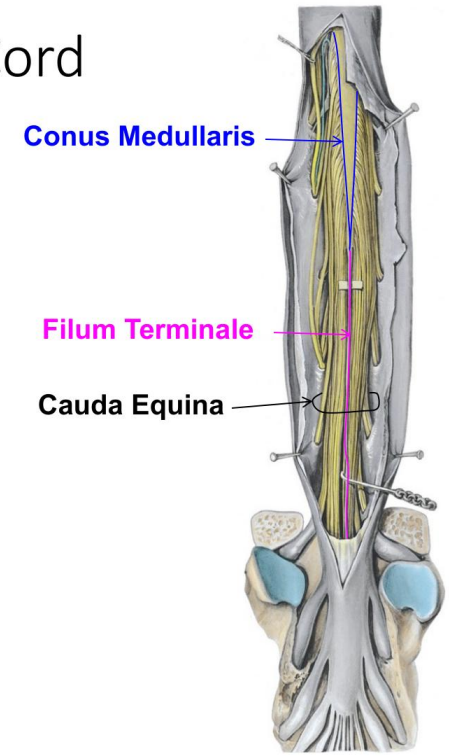
a. Part of the **central nervous system (CNS)** and continuous with the brainstem (medulla). Major reflex center and conduction pathway between the body and the brain
b. Housed within the vertebral column, it extends from the foramen magnum to the **L1-L2 vertebral level** in adults
c. Diameter not uniform
    i. **Cervical Enlargement** – C4-T1 spinal cord levels; associated with the brachial plexus, which innervates the entire upper limb
    ii. **Lumbosacral Enlargement** – L1-S3 spinal cord levels; associated with lumbar and sacral plexuses, which innervates the entire lower limb
    iii. **Conus Medullaris** – tapering inferior end of the spinal cord that terminates inferior to the lumbar enlargement
    iv. **Filum Terminale Internum** – continuation of the pia mater; extends as slender, anchoring filament from the tip of the conus medullaris to the coccyx

Spinal Nerve Organization:

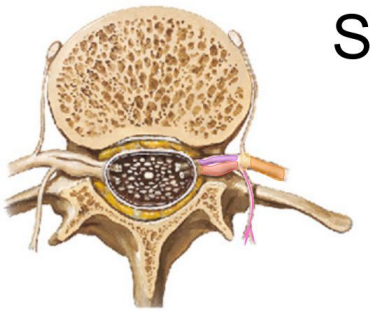
a. 31 pairs of spinal nerves:
    i. 8 cervical
    ii. 12 thoracic
    iii. 5 lumbar
    iv. 5 sacral
    v. 1 coccygeal

Spinal Cord



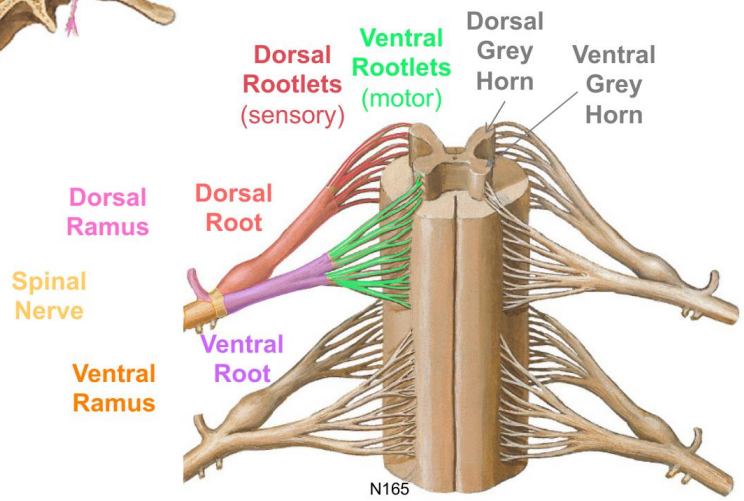


Spinal Cord



Dorsal Rami:
- Supply skin & deep muscles of back

Ventral Rami:
- Supply limbs & rest of trunk



In adults, the spinal cord is shorter than the vertebral column. Originally in the embryo, the spinal cord extended the full length of the spinal canal, so spinal cord segments were located at the level of the corresponding vertebra. However, growth of the spinal cord does not keep pace with the growth of the vertebral column, such that the spinal cord eventually only occupies the superior 2/3 of the spinal canal.

1. **Cervical segments** remain close to their corresponding vertebrae. **C1-C7 spinal nerves** exit superior to their corresponding vertebrae, while the C8 spinal nerve exits below the C7 vertebrae
2. Below the cervical region, spinal nerves exit the vertebral column by passing inferior to the pedicles of the corresponding vertebrae
3. The distance between a spinal cord segment giving rise to a pair of spinal nerves and the intervertebral foramen through which the nerves must exit the vertebral column increases as the cord descends
4. While the spinal cord ends at ~L1/2 in adults, nerve roots that branch off the lower lumbar and sacral regions of the spinal cord must still descend to exit the ***vertebral column at their respective intervertebral foramina***
5. The nerve roots continuing past the termination of the spinal cord comprise the cauda equina, which means “horse’s tail” due to the appearance of all the nerve roots extending inferiorly

Multiple **rootlets** emerge from ventral (anterior) and dorsal (ventral) sides of the spinal cord that merge to form:

1. **Ventral roots** of spinal nerves
    a. Contain **motor** (efferent) **fibres** and **presynaptic autonomic fibres**
    b. Cell bodies of axons contributing to the ventral roots are located within the **anterior horn of the gray matter** within the spinal cord
2. **Dorsal roots** of spinal nerves
    a. Contain **sensory** (afferent) **fibres**
    b. Cell bodies of axons contributing to dorsal roots are located in the **spinal ganglia** (dorsal root ganglia) that appears as a small swelling on dorsal root

Ventral and dorsal roots unite to form a mixed (carries sensory and motor innervation) **spinal nerve** within the **intervertebral foramen**

Each spinal nerve divides into two rami:

1. **Dorsal Ramus** – supplies the skin and deep muscles of the back
2. **Ventral Ramus** – supplies the limbs and the rest of the trunk

Clinical Note: The lumbar spinal nerves increase in size from superior to inferior, while the lumbar intervertebral foramina decrease in diameter. Thus, the more inferior nerves are more vulnerable to compression (L5 spinal nerves are the largest, while their intervertebral foramina are the smallest).

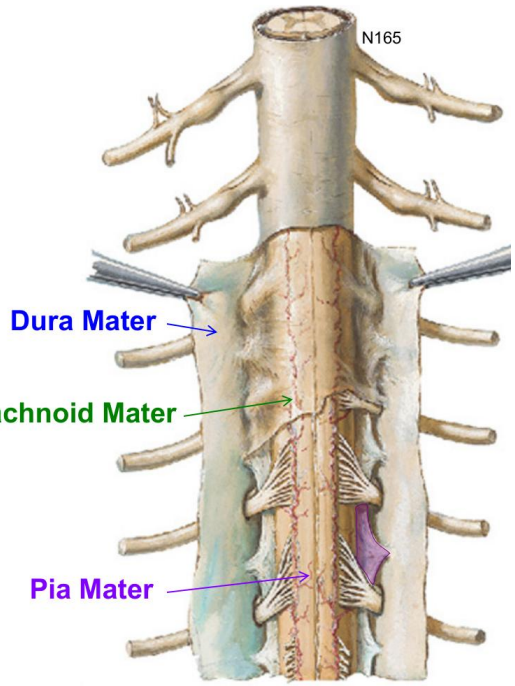
Each pair of spinal nerves represents a specific segment of the spinal cord and provides sensory innervation for a specific area of skin (**dermatome**) and motor innervation to a specific group of muscles (**myotome**)

Spinal Meninges:

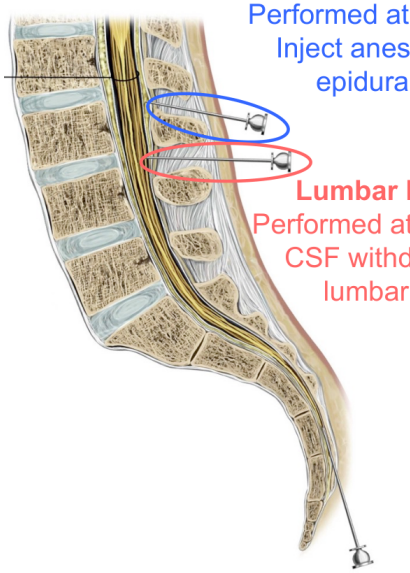
a. Membranes that surround, support, and protect the spinal cord and spinal nerve roots
b. Three layers:
   - **i. Spinal Dura Mater ("tough mother")**
     1. Tough outermost covering of spinal cord
     2. Forms **spinal dural sac**, a long tubular sheath within vertebral canal
        a. Extends inferiorly to the S2 vertebral level
        b. Pierced by spinal nerves
        c. Anchored to the coccyx by the **filum terminale externum**
        d. Continuous with cranial dura mater at foramen magnum
     3. **Epidural Space** – actual space superficial to dura mater; occupied by fat and the internal vertebral venous plexus
     4. **Subdural Space** – potential space between the dura and arachnoid mater
     5. **Clinical Note:** During an **epidural block,** anesthetic can be injected into the epidural space, which would directly affect the spinal nerve roots of the cauda equina after they exit from the dural sac. The patient would lose sensation inferior to the level of the block. This is frequently used during labor and delivery.
   - **ii. Spinal Arachnoid Mater**
     1. Delicate, avascular membrane (middle layer)
     2. Encloses **subarachnoid space** containing **cerebrospinal fluid** (CSF) and vasculature to the spinal cord
     3. Not attached to dura mater, but held against it by pressure from the CSF
     4. **Clinical Note:** A **lumbar spinal puncture** is performed to obtain a CSF sample from the **lumbar cistern**, an enlargement of the subarachnoid space between the termination of the spinal cord (L2 level) and termination of the dural sac (S2 level) that houses the cauda equina. A needle is inserted into the subarachnoid space while the patient is lying on his/her side with the trunk flexed. It is usually performed at the L3/L4 or L4/L5 vertebral level.
   - **iii. Spinal Pia Mater**
     1. Innermost covering membrane directly covering the spinal cord, nerve roots, and vessels
     2. **Denticulate ligaments** – tooth-like extensions extend between anterior and posterior roots that anchor the spinal cord in the dural sac
     3. Continues as **filum terminale internum** inferior to the conus medullaris

Meninges

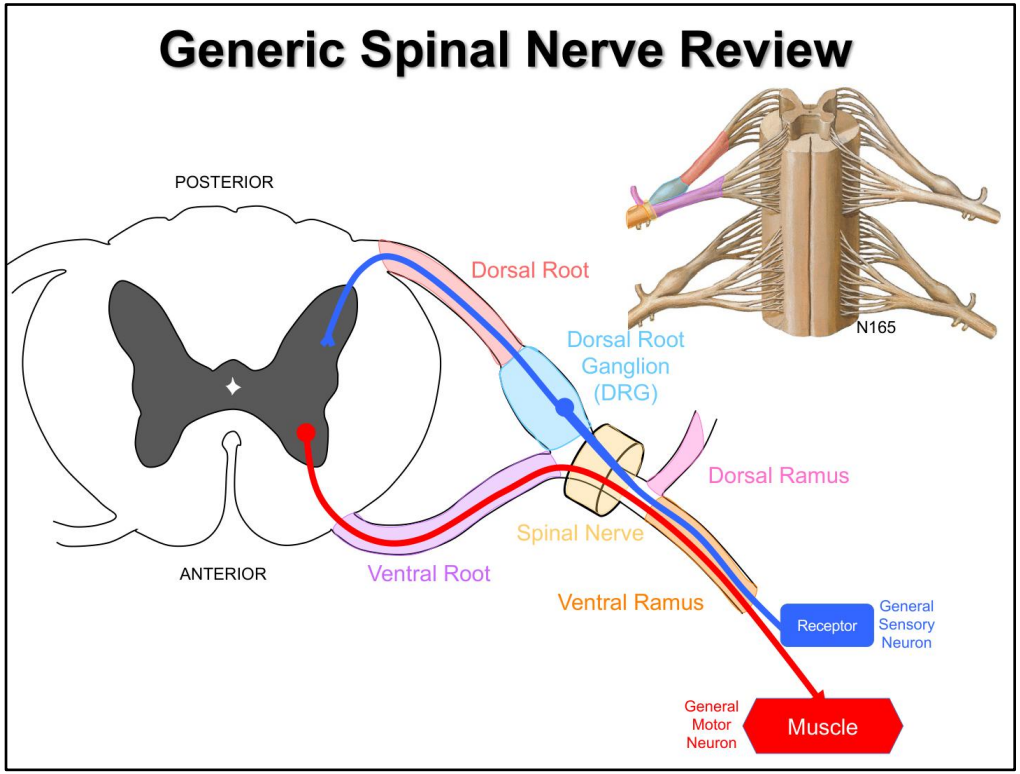
- Continuous with the meninges of the brain
- Surround, support, & protect
- **Dura Mater** "tough mother"
  - Forms dural sac (to S2)
- **Arachnoid Mater** "web like"
  - Delicate, avascular layer
- **Pia Mater:**
  - Innermost layer
  - **Denticulate ligaments** anchor spinal cord
  - Continues as filum terminale



Meninges

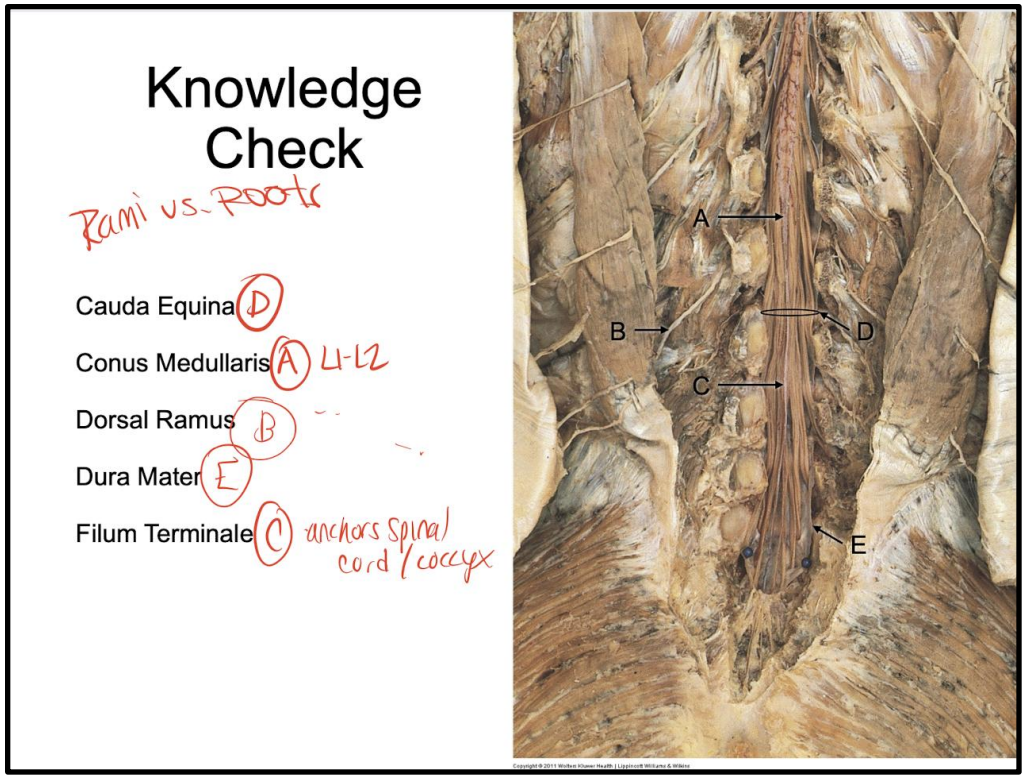


- **Epidural Space**
  - Real space
  - Superficial to dura
  - Occupied by fat & venous plexus
- **Subdural Space**
  - Potential space
  - Between dura & arachnoid
- **Subarachnoid Space**
  - Real space
  - Contains **cerebrospinal fluid (CSF)** & vasculature



Somatic Pathway:

a. **General Sensory (Afferent) Neuron** - Transmits sensations from a receptor on the body wall through the spinal nerve and dorsal root to the CNS. Cell bodies for these neurons are located within the dorsal root ganglia (DRG).
b. **Synapse** – Once inside the spinal cord, the sensory fibres enter the dorsal grey horns where they will synapse with a secondary neuron, which will transmit the impulse to the brain.
c. **General Motor (Efferent) Neuron** – leaves the spinal cord through the **ventral root**, enters the **spinal nerve** and passes along motor branches of either the **anterior** or **posterior primary rami** to reach its target (skeletal muscles). Cell bodies for these neurons are located within the ventral grey horn of the spinal cord.
    i. **Reflex Arc** – sensory fibres may either pass directly to the ventral grey horn or they synapse with an interneuron, which passes to the ventral grey horn. In this situation, the sensory neuron synapses directly with the motor neuron, bypassing the brain.



Additional Notes from Text Notes:

MSK Development Review:

A. As the notochord and neural tube form in the third week, the intraembryonic mesoderm lateral to these structures thickens to form two longitudinal columns of paraxial mesoderm

B. By the end of the third week the paraxial mesoderm differentiates, condenses and begins to divide into blocks of paired mesoderm, called somites.

C. Muscular System:

a. The **myotome** portion of the **dermomyotome** gives rise to the muscles of the axial skeleton, body wall, and limbs
b. By the end of the fifth week, each typical myotome divides into a dorsal **epimere (epaxial division)** and a ventral **hypomere (hypaxial division)**

i. **Myoblasts** from the **epimere** form the **epaxial muscles.**
    1. Splenius muscles, erecter spinae muscles, and the transversospinalis muscle group
ii. **Myoblasts** from the **hypomere** form the **hypaxial muscles.**
    1. Lateral and ventral muscle of the trunk and all of the muscles of the limbs

c. Derivatives of each **somite** are supplied by the nerve that originally served that s**omite**. Thus, as muscle differentiate into epimeres and hypomeres, each spinal nerve divides into a **dorsal primary ramus**, to serve **epimere-derived** muscles from that segment, and a **ventral primary ramus** to innervate **hypomere-derived** muscles from that segment
    i. Some muscles, such as the intercostal muscles, remain segmentally arranged but most myotomes fusewith others to form individual skeletal muscles; thus, most muscles are innervated by more than one spinal segment
        1. This occurs by innervation to a muscle from multiple spinal nerves (i.e. muscles of the trunk) or the joining of multiple spinal nerves into single nerves in the brachial and lumbosacral plexuses for limb muscle innervation

D. Cutaneous Innervation:

1. Recall, each somite is served by a single spinal nerve
2. During the fifth week, **sensory peripheral nerves** grow from their developing limb plexuses (brachial and lumbosacral), using the motor nerves as their 'guides' into the mesenchyme of the limbs
3. The spinal nerves are distributed in segmental bands called **dermatomes.** A **dermatome** is the area of skin supplied by a single spinal nerve and its dorsal ganglion
    a. However, dermatomes considerably overlap one another
4. Even when muscle cells migrate, within the elongating limb, they continue to be innervated by the spinal nerve from their segment of origin
    a. This explains the **segmental pattern** of innervation of the muscles (myotomes) and the skin **(dermatomes)**
    b. Note there is minimal dermatome overlap where non-sequential spinal segments are adjacent at the **ventral axial line**

Non-Mechanical Axial Pain

- Neurological edition: Problems to the side -radiculopathy
    - Problems to the side - radiculopathy
    - Problems to the centre
        - LUMBAR (RED FLAGS)
        - CERVICAL (RED FLAGS)
    - Everything Else - RED FLAGS
        - Inflammatory
        - Fractures
        - Visceral
        - Infection
        - Neoplasm
    - **Done by Dr. Caitlin Cassidy**

Non-Mechanical Spine Pain (Neurology Edition) Part 1: Problems to the Side

Road Map!

- What's the Big Deal
- Mechanical Axial Pain
  - Bones/ligaments/joints
    - Disc
    - Facet
  - Muscles
- Non-Mechanical Axial Pain
  - Neurology edition
    - Problems to the side – radiculopathy
    - Problems to the centre
      - Lumbar (RED FLAGS)
      - Cervical (RED FLAGS)
  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Non-Mechanical Neurologic Neck/Back Pain

RADICULOPATHY

- "Radix" = "root"
- "Pathos" = "suffering/disease"
- Results from compression/malfunction of the spinal nerve roots in or around the neural foramen of ANY CAUSE
  - Not just degenerative causes (including disc herniation) though those are BY FAR most common
    - Others: neoplasm (primary, mets, carcinomatous meningitis), infectious (abscess, TB, etc), sarcoidosis, diabetic neuropathy affecting root, trauma, AVM

Non-Mechanical Neurologic Neck/Back Pain

RADICULOPATHY

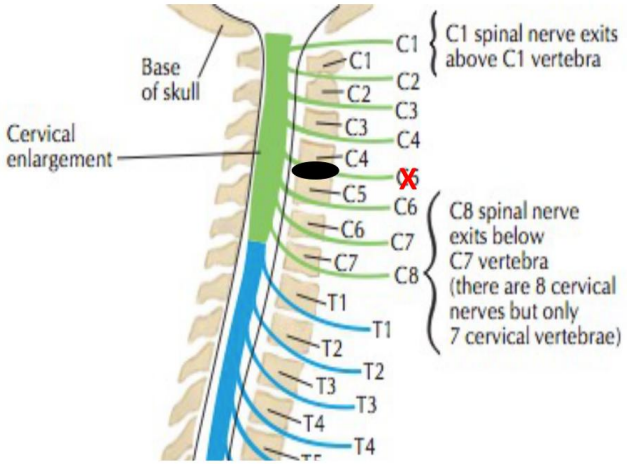
- O – may be acute or insidious depending on cause
- L – arms/legs > neck/back
  - Dermatomal distribution of pain, sensory changes
- D – variable
- C – burning/tingling, electrical
- A – alleviated by positions that relieve nerve tension (ie. arm overhead, legs curled); associated with weakness/hyporeflexia in a myotomal distribution
- R – following dermatomal pattern
- S - variable
- Red flags

Radiculopathy

- Lumbar spine
  - <1% caused by something other than degenerative changes
  - Most commonly L5 and S1
- Cervical spine
  - Most commonly C7, C6
  - Most commonly related to cervical spondylosis

Spinal Nerves

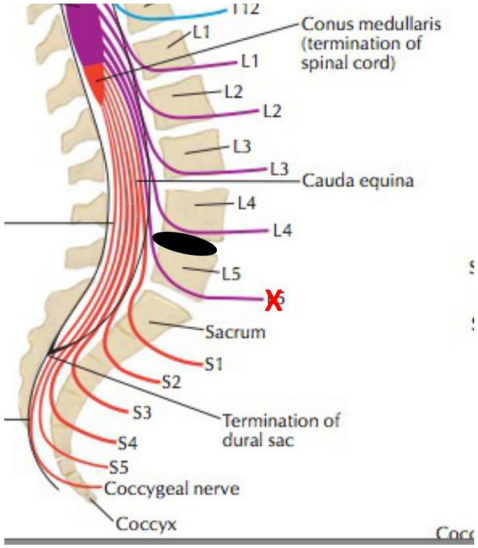
- Cervical nerves exit **above** corresponding vertebra
  - EXCEPT C8
  - So if a disc bulged here….
  - Which root would be impacted?
- Cervical disc herniations affect nerve of LOWER vertebra
  - Ie. C4/5 herniation affects C5



Non-Mechanical (Neurology)

Spinal Nerves

- All nerves below C7 exit **below** corresponding vertebra
  - So if a disc bulged in the lumbar spine….
  - Which root would be impacted?
  - Lumbar disc herniations affect nerve of LOWER vertebra
    - Ie. L4/5 herniation affects L5
      - Because of the angle of nerve travel in the L-spine, the L4 root has already exited
      - As a result, the herniation affects the "traversing" root



For your reading pleasure

	Dermatome	Myotome	Reflex
C5	Lateral shoulder to lateral elbow	Shoulder abduction, elbow flexion	Biceps (C5/6) Brachioradialis (C5/6)
C6	Lateral forearm, digits I and II	Elbow flexion, wrist extension	Biceps (C5/6) Brachioradialis (C5/6)
C7	Digit III	Elbow extension	Triceps (C7)
L4	Medial lower leg, medial malleolus	Dorsiflexion	Patellar (L4)
L5	Dorsum of foot (1st- 3rd webspace)	Great toe extension	Hamstrings (L5) (often absent "normally," be sure to check contralateral response)
S1	Lateral foot, lateral heel	Plantar flexion	Achilles (S1)

Non-Mechanical (Neurology)

Radiculopathy Management

- Shockingly, not that different from muscular/disc/facet
- Conservative management
  - Rule out red flags - if any concern, do appropriate imaging
    - Imaging also appropriate if neurologic impact of radic is severe or progressing (ie. significant arm weakness)

What Would be Appropriate Imaging?

- Remember, nerve compression can be from ANYTHING
  - Degenerative, tumours, abscesses, spinal instability
- Canadian Association of Radiology Guidelines

(apply in cases where you've considered red flags and found none…if any red flags, do appropriate imaging)

C04. Neck pain, brachialgia, degenerative change	MRI	Indicated [B]	0	Imaging is only indicated when there are neurological signs or symptoms, or if pain persists after conservative management for more than four weeks.
	CT	Indicated only in specific circumstances [B]	☢☢	CT is indicated only when MRI is contraindicated or not available.
	XR	Not Indicated [B]	☢	Degenerative changes begin to appear on XR in early middle age and are usually unrelated to the patient's symptoms.

https://car.ca/wp-content/uploads/Spine.pdf

Non-Mechanical (Neurology)

Radiculopathy Management

- Shockingly, initial approach not that different from muscular/disc/facet
- Conservative management
  - Rule out red flags - if any concern, do appropriate imaging
    - Imaging also appropriate if neurologic impact of radic is severe or progressing (ie. significant arm weakness)
    - Can also consider EMG to test function of the nerve root
  - Education
    - Reassure
    - Discuss excellent prognosis (most resolve spontaneously over months)
  - Relative rest
  - Modalities (ice, heat)
  - Topicals
  - Oral anti-inflammatories if necessary
    - Short course opioids are appropriate if pain is severe
  - Physiotherapy
    - Strengthen postural support

Radiculopathy Management

- Next steps
  - Interventional options - epidural steroid injections
  - Surgical options – decompression
    - Refractory symptoms despite adequate trial of conservative treatment
    - Progressive or severe neurological deficit
    - Certain red flag conditions (ie. tumour, large abscess, trauma)

Non-Mechanical Spine Pain (Neurology Edition) Part 2: Problems to the Centre

Road Map!

- What's the Big Deal
- Mechanical Axial Pain
  - Bones/ligaments/joints
    - Disc
    - Facet
  - Muscles
- Non-Mechanical Axial Pain
  - Neurology edition
    - Problems to the side – radiculopathy
    - Problems to the centre
      - Lumbar (RED FLAGS)
      - Cervical (RED FLAGS)
  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Location, Location, Location

- Central canal stenosis at different sites along the spine looks and is treated entirely differently…why?
- We'll discuss the L spine and the C spine separately

Non-Mechanical Neurologic Spine Pain

LUMBAR CENTRAL STENOSIS

- O – typically insidious
- L – legs > back
- D – variable
- C – burning/tingling, electrical
- A – lumbar stenosis aggravated by walking
- R – typically non specific, often distal>proximal extremity
- S - variable
- Red flags

Central Stenosis – Lumbar Spine

- Typically spinal cord ends at ~L1
  - SO, stenosis of the lumbar central canal = root problem
- Classic
  - Leg>back pain (though patients often describe both)
  - Typically distal parts of extremities more effected
    - Compression of vascular supply to the nervous structures
  - Symptoms worse with walking certain distance (CLAUDICATION)
    - Relief with forward flexion
      - Relief with using grocery cart or walking uphill
    - Relief with sitting down (but not within minutes like vascular)
  - No bowel or bladder problems

Non-Mechanical (Neurology)

Central Stenosis – Lumbar Spine

- Cauda Equina Syndrome
  - Further down severity spectrum of central canal stenosis in L spine
  - Greater deal of compression (from whatever cause) leads to significant neural compromise, particularly at lower root levels
    - Weakness
    - Saddle anaesthesia
    - Bladder changes (urinary retention)
    - Bowel changes (fecal incontinence)

RED FLAGS!!!



Central Stenosis – Lumbar Spine

- Management
  - Imaging decisions are based on presentation
    - LBP with mild to moderate claudication – consider no imaging or non-urgent MRI (particularly if patient is a potential surgical candidate)
    - Symptoms of cauda equina – urgent MRI
  - Conservative management for those with "manageable" symptoms
    - Flexion based activities and physiotherapy is mainstay
  - Cauda equina syndrome is a neurosurgical emergency
    - Outcomes are significantly better if decompressive surgery performed within first 48 hrs after onset of symptoms

Non-Mechanical (Neurology)

Central Stenosis – Cervical Spine

- No way to avoid the spinal cord up here therefore central compression that's significant enough = cord compression
  - Medical term for this = cervical myelopathy
- Clinical picture of atraumatic spinal cord injury
  - Decrease in upper extremity dexterity and wasting in hand intrinsics
  - Upper extremity paraesthesiae (distal>proximal)
  - Urinary incontinence (may be due to retention and overflow)
  - Gait changes/balance problems
  - Often concurrent c/o features of radiculopathy at particular level
    - Hard to squish the cord without getting a root at the same time!
  - Upper motor neuron signs below the level of the lesion

Central Stenosis – Cervical Spine

- MRI is imaging modality of choice
- Management
  - Requires urgent neurosurgical/spine consultation for consideration of decompression

Non-Mechanical Spine Pain ("Everything Else")

Road Map!

- What's the Big Deal
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- Non-Mechanical Axial Pain
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  - "Everything else" – RED FLAGS
    - Inflammatory
    - Fractures
    - Visceral
    - Infection
    - Neoplasm

Inflammatory Spine Pain

- More on this to come in Rheumatology lectures
- When axial skeleton pain is truly "inflammatory"
  - Pain IMPROVES with movement (opp of classic mech. spine pain)
  - Often involves SI joints
  - Night pain can be a feature, usually in 2nd half of night
  - Significant AM stiffness
  - Consider important features on ROS – eye symptoms, rashes, family history, enthesopathies, other inflammatory joint pain
- Think of seronegative sponyloarthropathies (ankylosing spondylitis, psoriatic arthritis, IBD associated arthritis, reactive arthritis)

Fractures

- Osteoporotic vertebral compression fractures
  - Axial pain with radiation
  - May follow minor trauma or no clear inciting incident
  - Think of this in elderly, patients on steroids, etc
  - Pain management, treat the osteoporosis!



Fractures

- Spondylolysis
  - Fracture of pars interarticularis
  - Think of this in gymnasts, football players, weight lifters and other active young people who do a lot of extension of the back
  - May result in spondylolisthesis



Visceral Back/Neck Pain

- Cardiovascular
  - AAA – pulsatile mass
  - Vertebral insufficiency – dizziness, cranial nerve findings
  - Atypical cardiac pain
- GI
  - PUD
  - Pancreatitis
  - Others
- GU
  - Pyelonephritis
  - Nephrolithiasis
  - Gynecologic causes (ie. Endometriosis)

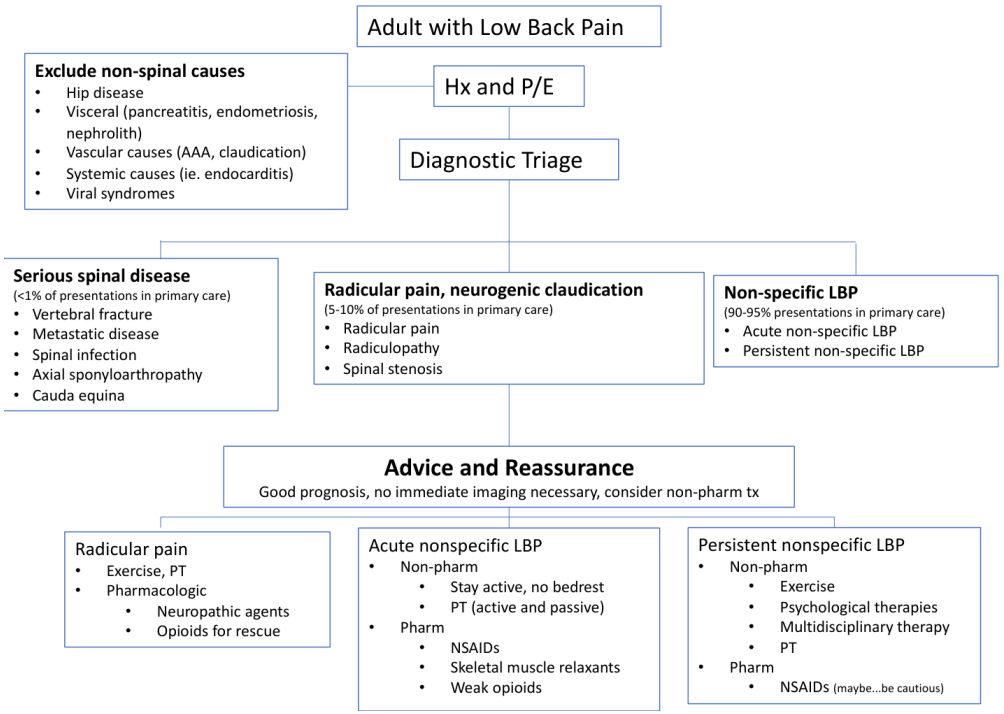
Other Bad Things

- Infection
  - Osteomyelitis, discitis, epidural abscess
    - Infection red flags: fevers, sweats, history of IV drug use, history of immunosuppression
- Neoplasm
  - Bone tumour, epidural or intradural mass, drop metastases, referred pain (ie. Renal carcinoma)
    - Neoplasm red flags: young age, severe night pain, unintentional weight loss, night sweats, history of cancer, associated signs (ie. hematuria)

How To Manage This???

- OLDCARS
  - Any red flags should lead to appropriate investigations (ie. Plain films if worried about fracture, MRI if worried about significant neurologic compromise/infection/neoplasm, etc)
  - In the absence of red flags, initial management of mechanical/nonspecific back pain, AND of non-emergent neurologic types of back pain (ie. radiculopathy, lumbar stenosis) is similar so start there!

Adapted from Traeger et al, 2017



Summary

- Recognize and identify common clinical spine problems
  - The “usual:” mechanical/nonspecific pain (ie. disc, facet), radiculopathy, non-cauda equina spinal stenosis
- Describe the anatomical basis of common spine disorders
- Know red flags and their significance (ie. what clinical entities they suggest)
  - Neoplasms, inflammatory pain, infections, fractures, urgent/emergency neurologic conditions
- Develop an approach to management of spinal conditions

A Few References

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